Introduktion til AI

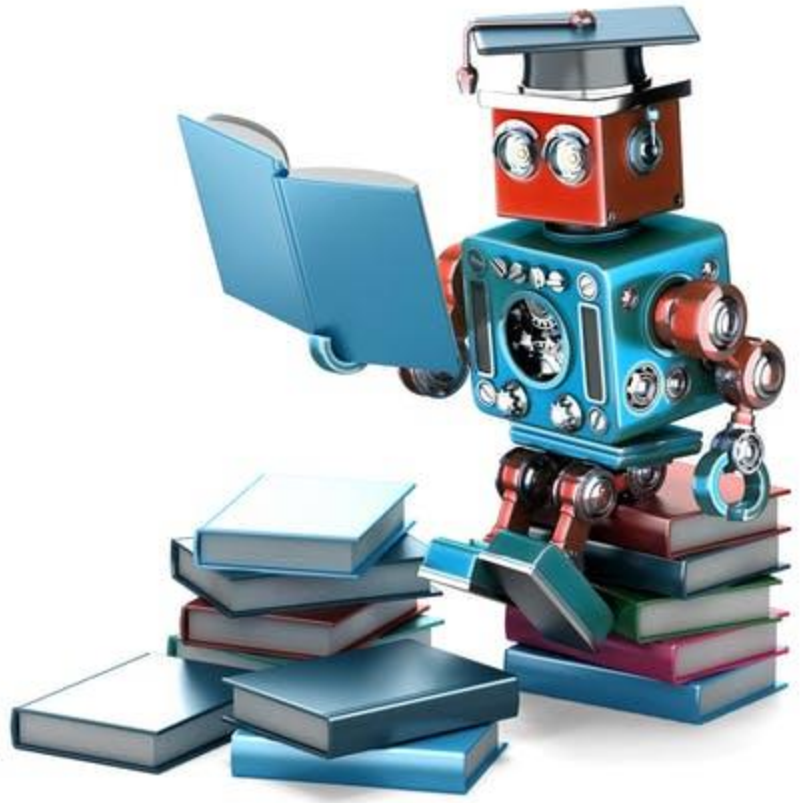


Professor Lasse Riis Østergaard, AAU
Dept. of Health Science and Technology
Medical Informatics and Image Analysis

AALBORG UNIVERSITY
DENMARK

Agenda

- Introduktion til kurset
- Hvad er machine learning og AI?
- Opgaver



Kursusbeskrivelse

Dette kursus giver grundlæggende viden og færdigheder i kunstig intelligens med fokus på maskinlæringsteknikker. Kurset giver studerende en forståelse for grundlæggende begreber inden for AI og udstyrer dem med værktøjer til at arbejde med maskinlæringssystemer. Gennem en kombination af teori og praktiske øvelser vil studerende lære at anvende AI-teknikker til forskellige opgaver.

I kurset skal vi bla. lære de grundlæggende koncepter inden for AI og maskinlæring, og forstå forskellen mellem supervised og unsupervised læring. Et vigtigt element er også at ***komme igennem hele maskinlærings-processen fra dataforberedelse til modeltræning, finjustering og vedligeholdelse af det færdige system***.

Gennem en kombination af teori og praktiske øvelser vil der opbygges en grundviden inden for AI og ML som fundament til at anvende denne viden i både forskning og industrielle kontekster.

Studieordning

INTRODUKTION TIL AI **2023/2024**

MODULETS INDHOLD, FORLØB OG PÆDAGOGIK

Kursets mål er at introducere de studerende til fundamentale koncepter og modeller indenfor kunstig intelligens samt relevante matematiske koncepter. Kurset indeholder således en beskrivelse af hvad features er, hvordan de udspænder featurerum og hvordan man matematisk kan modellere og genkende forskellige mønstre i sådanne rum. Ydermere introducerer kurset systematiske metoder til evaluering af AI modeller.

LÆRINGSMÅL

VIDEN

Studerende som gennemfører modulet vil opnå:

- Viden om basale elementer og opbygning af et AI system
- Viden om forskellige typer af features som forklarende variable
- Kendskab til forskellige kontekster og scenarier hvori AI indgår

FÆRDIGHEDER

Studerende som gennemfører modulet vil være i stand til at:

- Anvende og evaluere basale AI modeller
- Visualisere, analysere og udvælge features
- Udføre relevante operationer med vektorer

KOMPETENCER

Studerende som gennemfører modulet vil være i stand til at:

- Omsætte viden om features og basale AI modeller til design, udvikling og evaluering af et simpelt AI system
- Evaluere anvendelsen af prædiktions-modeller kontra klassifikations-modeller

UNDERVISNINGSFORM

Se oversigt over undervisningsformer under §17.

EKSAMEN

FORUDSÆTNING FOR INDSTILLING TIL PRØVEN

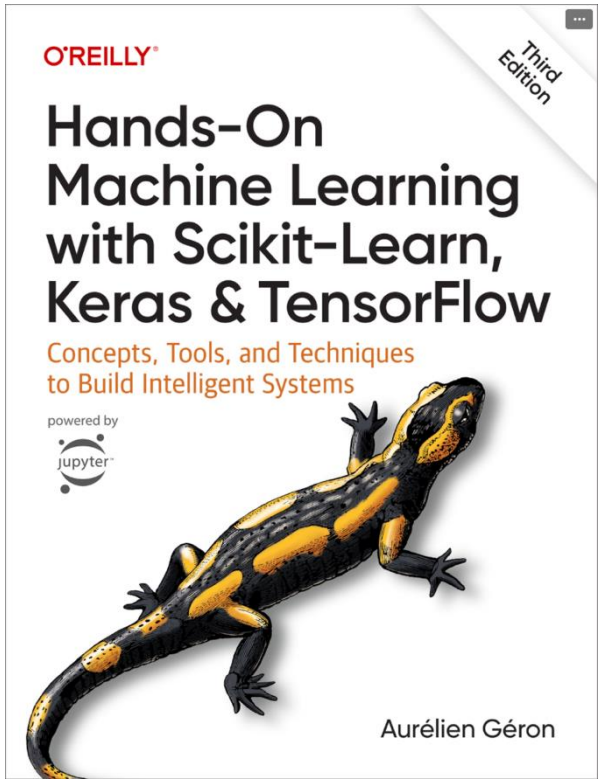
- For at kvalificere sig til eksamen, skal den studerende have afleveret obligatoriske opgaver rettidigt.

PRØVER

Prøvens navn	Introduktion til AI
Prøveform	Skriftlig eller mundtlig
ECTS	5
Tilladte hjælpemidler	Med visse hjælpemidler: Se semesterbeskrivelse
Bedømmelsesform	Bestået/ikke bestået
Censur	Intern prøve
Vurderingskriterier	Vurderingskriterierne er angivet i Universitetets eksamensordning

Læsestof

Vi kommer til at bruge bogen Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 3rd Edition af Aurélien Géron. Køb e-bog eller fysisk bog alt efter hvad du foretrækker.



Software

- Programming language: Python
- Integrated development environment (IDE): Visual Studio Code (VS Code)
- Libraries: Scikit-Learn (se User Guide!), Numpy, Pandas, Matplotlib, Seaborn, …
- Open source web application: Jupyter Notebook

- Følg anvisninger fra Andreas i forbindelse med AI-Programmering og Thomas i forbindelse med Lektion 2

Prerequisites…not really

- Basic statistics
- Probability theory
- Some mathematical concepts
- Python programming
- Linear algebra

Undervisere

- Lasse Riis Østergaard



- Thomas Kronborg Larsen



- Anders Ulrich Eriksen



Lektioner

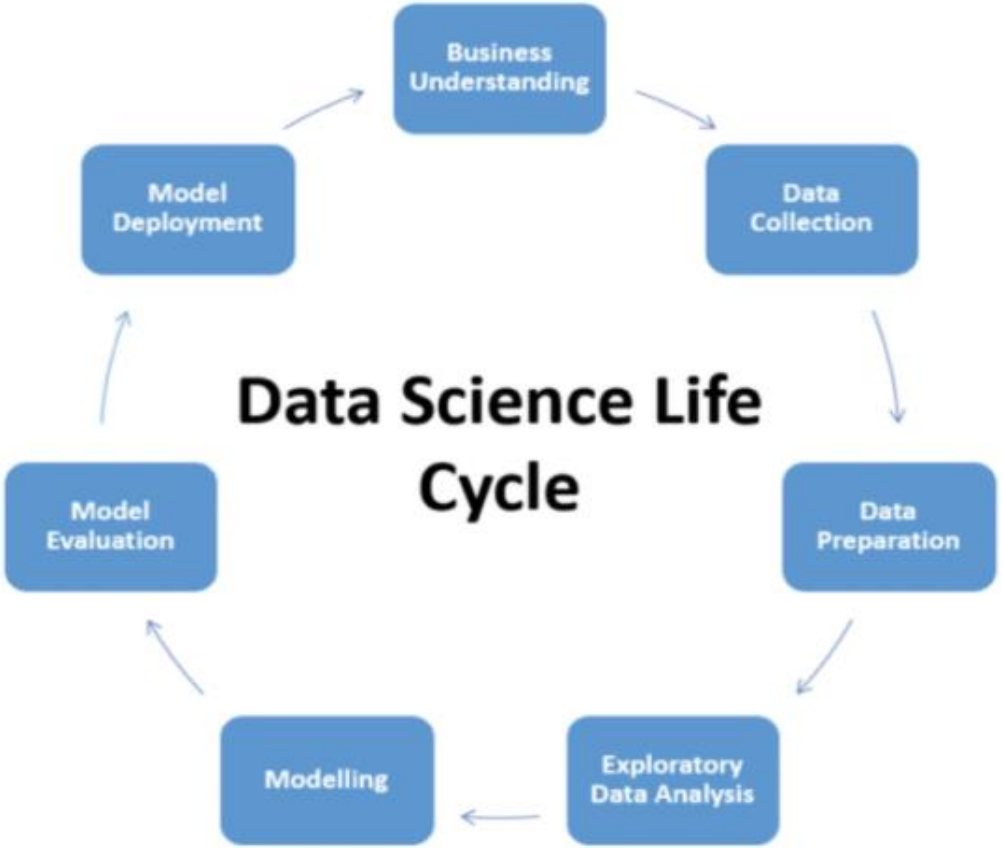
- Lektion 1: Introduction to AI
- Lektion 2: Python for Machine Learning + kNN
- Lektion 3: Main challenges of Machine Learning
- Lektion 4: End-to-End Machine Learning part 1
- Lektion 5: End-to-End Machine Learning part 2
- Lektion 6: Basic classification
- Lektion 7: Linear and logistic regression
- Lektion 8: Basic clustering
- Lektion 9: Decision trees and random forest
- Lektion 10: Examples from industry/research

Data science og ML/AI

- "Data science combines multiple fields, including statistics, scientific methods, artificial intelligence (AI), and data analysis, to extract value from data" - Oracle

- Indsamling af store og komplekse datasæt
- Frembringe viden og mening fra data
- Prædiktioner og beslutninger

The data science life cycle



Hvad er forskellen på AI, ML, deep learning og data science?

AI: enables the machine to think

- Understand, learn, making decisions and (causal) inferences

ML: statistical tools to explore and analyse data

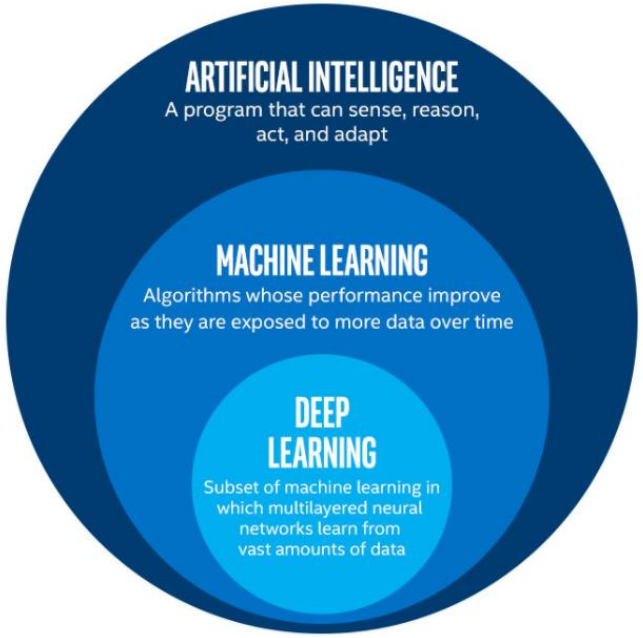
- Supervised, unsupervised, reinforcement

Deep learning: Multi neural network architecture

- ANN (e.g. numbers), CNN (e.g. images), RNN (e.g. time series)

Data science: all about data ☺

- data analysis, math, programming, IDE, data visualization, machine learning, web scraping, etc.



Why use Machine Learning?

- Explore/study problem
- Find patterns
- Write detection algorithm for each pattern
- Test program and repeat 1 and 2 until good enough to launch

- Long list of complex rules
- Hard to maintain
- P0

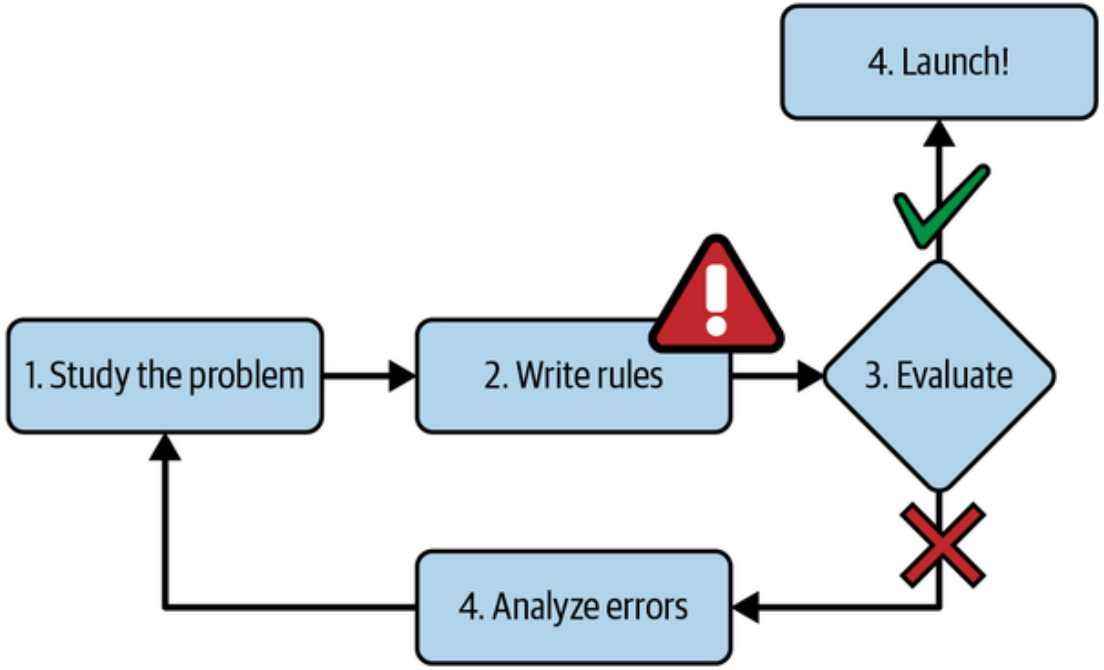


Figure 1-1. The traditional approach

Why use Machine Learning?

- Explore/study problem
- Collect training data
- Select features and model
- Test program and repeat 1 and 2 until good enough to launch

- Machine learning automatically learns patterns
- Much shorter program
- Easier to maintain
- More accurate
- PI

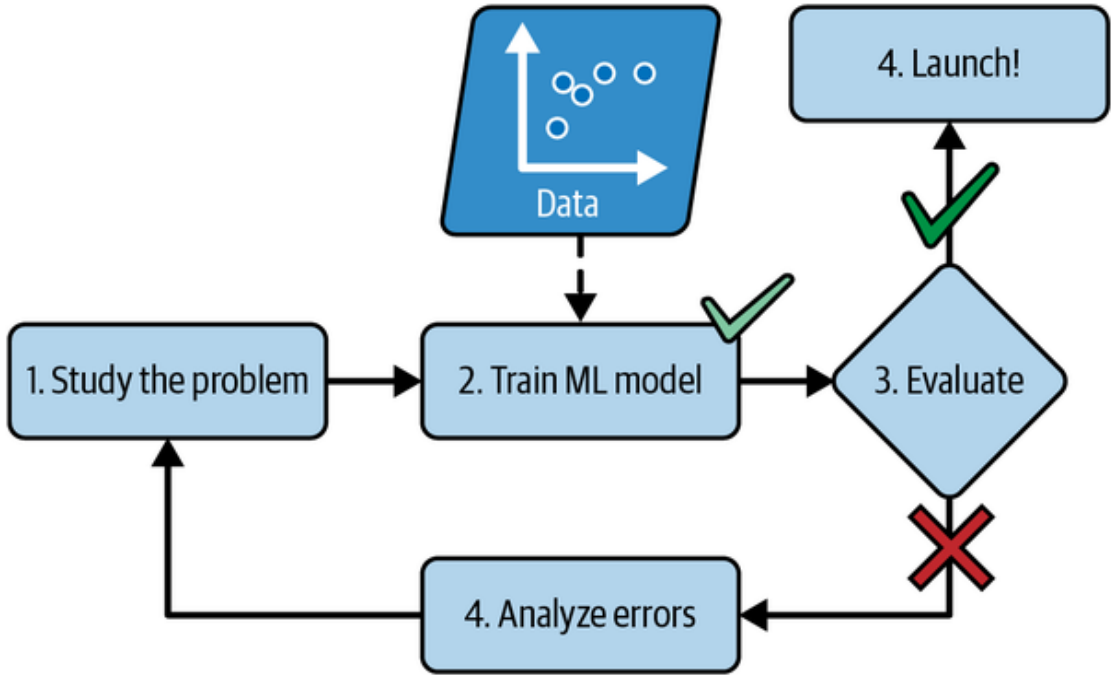
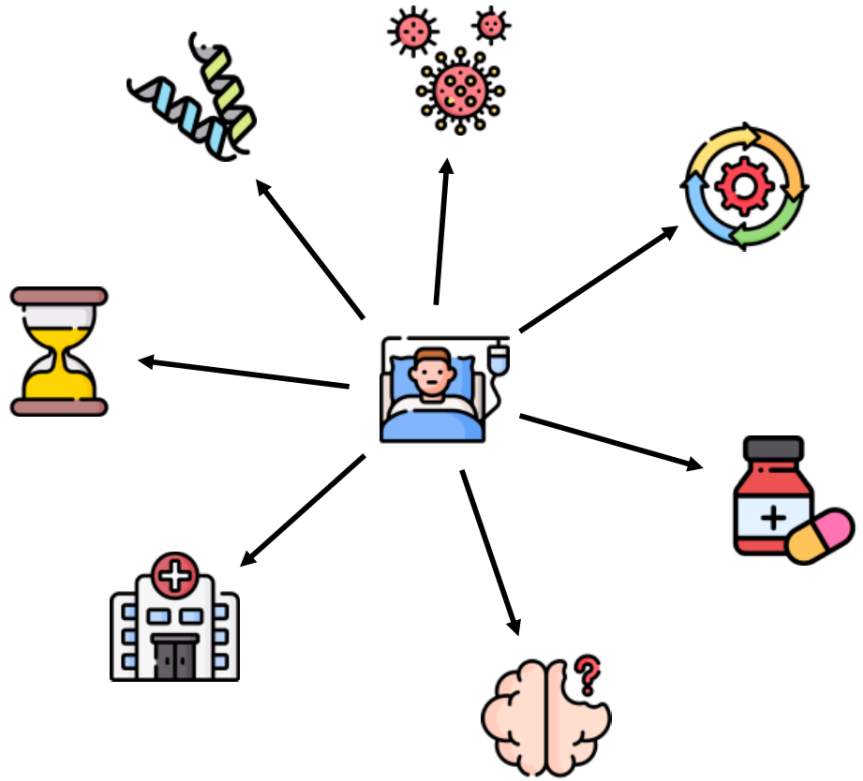


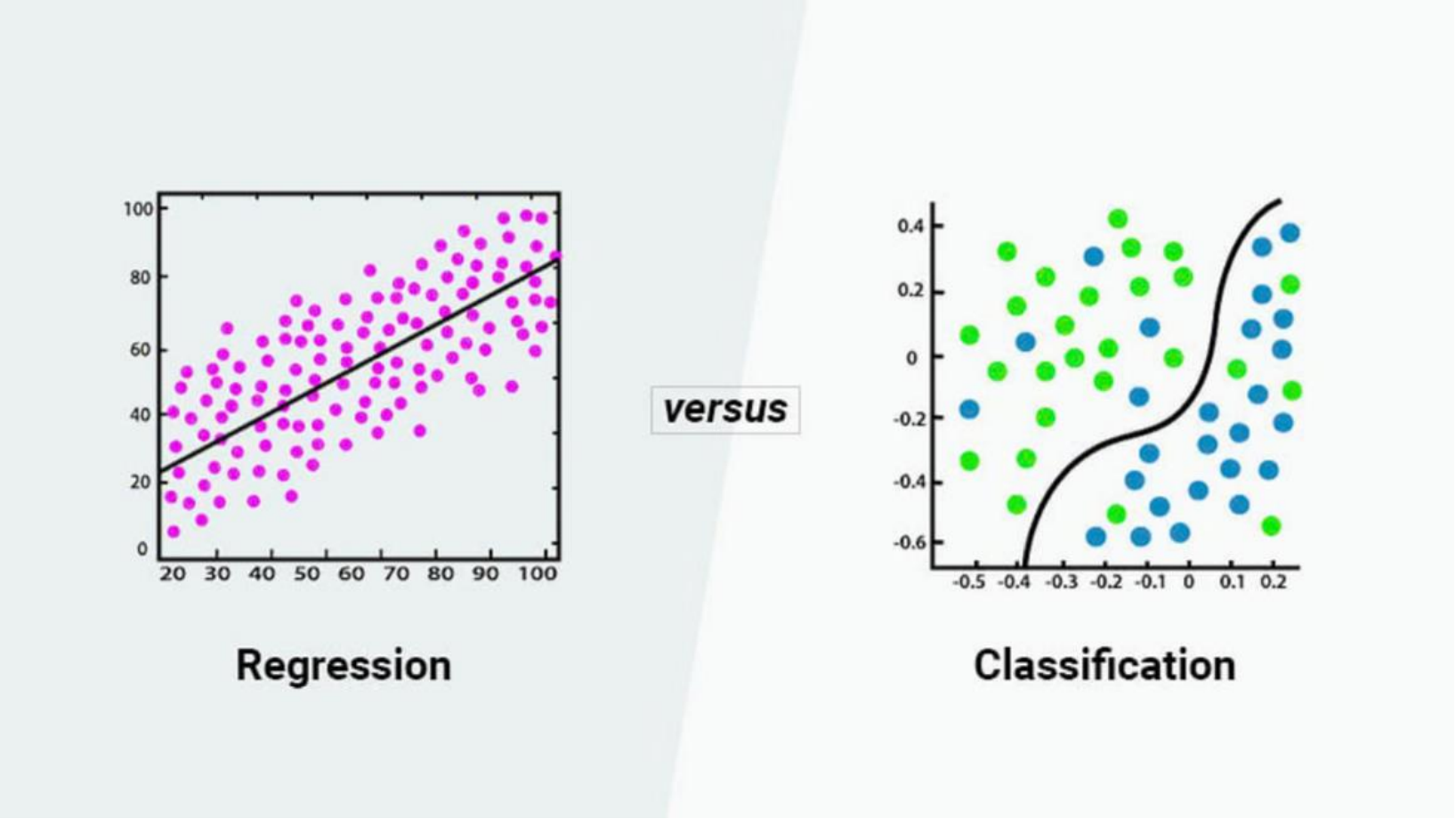
Figure 1-2. The machine learning approach

What tasks can Machine Learning solve?

- **Classification**
  - What conditions does a patient suffer from?
  - What is the next action we should perform?
  - Will the patient develop Alzheimer's disease?
  - What medication is the best to treat this patient?
- **Regression**
  - For how long will the patient be hospitalized?
  - How long will the patient survive?
  - What is the estimated CRP of the patient?



Regression vs Classification



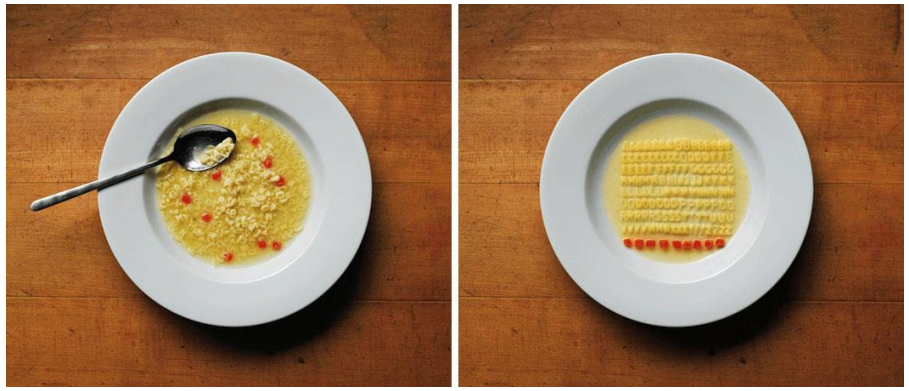
Pattern recognition

- Pattern recognition takes place when seeking to identify an ”object” based on observations characterising the ”object”.

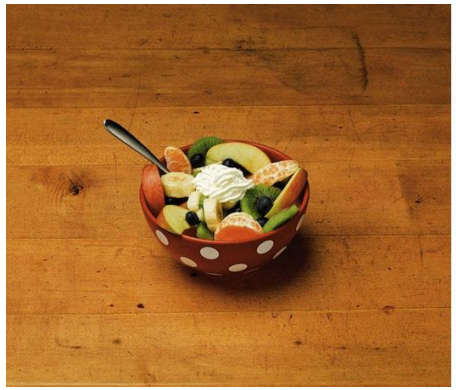
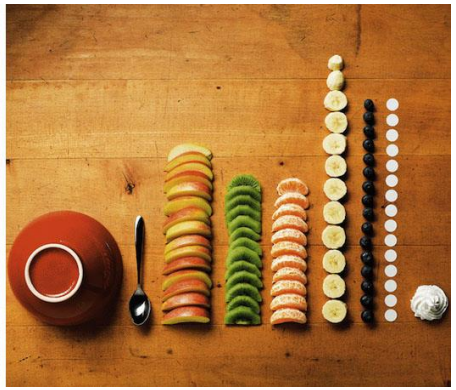
- Determine the category or class the ”object” belongs to.



”Tidying up” – finding patterns





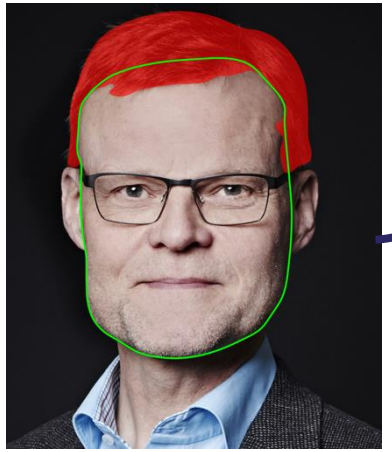



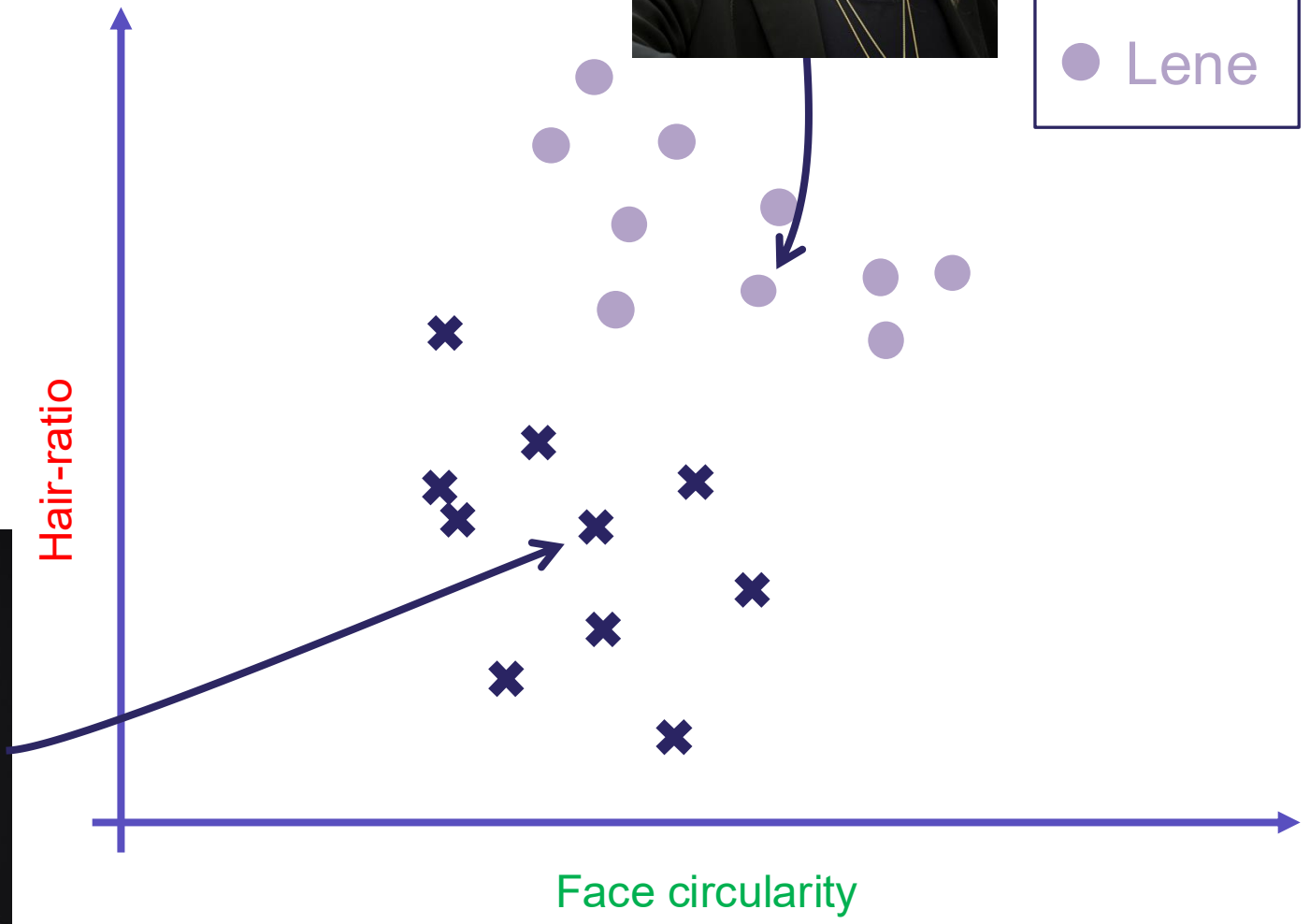


What is machine learning?

- Example: Differentiate between Per and Lene
  - Amount of hair?
  - Face circularity?





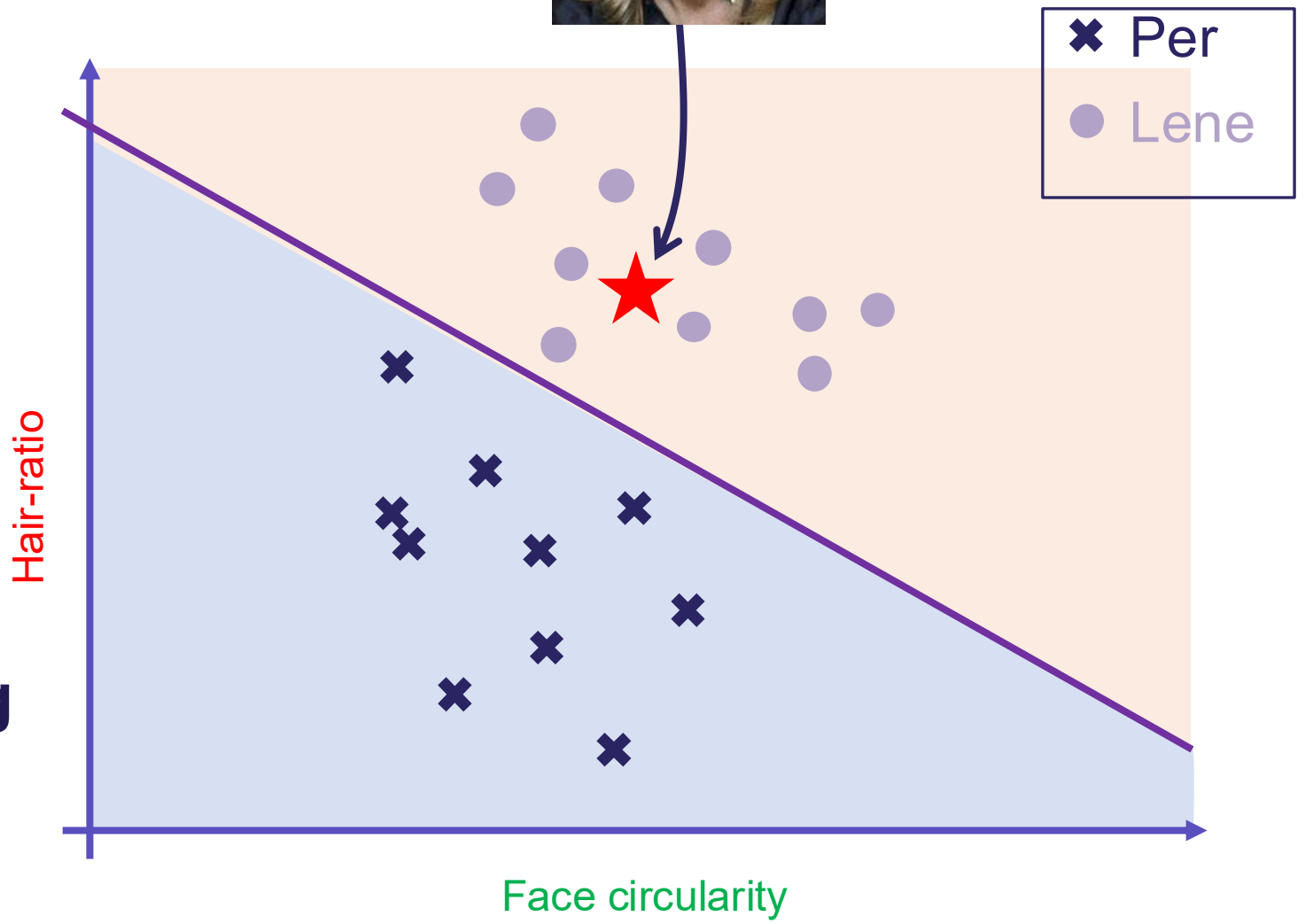


What is machine learning?



- New image – who?
  - Extract features
  - Where in feature space?
  - Compare to line
  - Decide who it is

- **This is Machine Learning**



Example



Sorting of fish on a conveyor belt

- Salmon or sea bass

Camera acquires images

Difference between the fish is identified

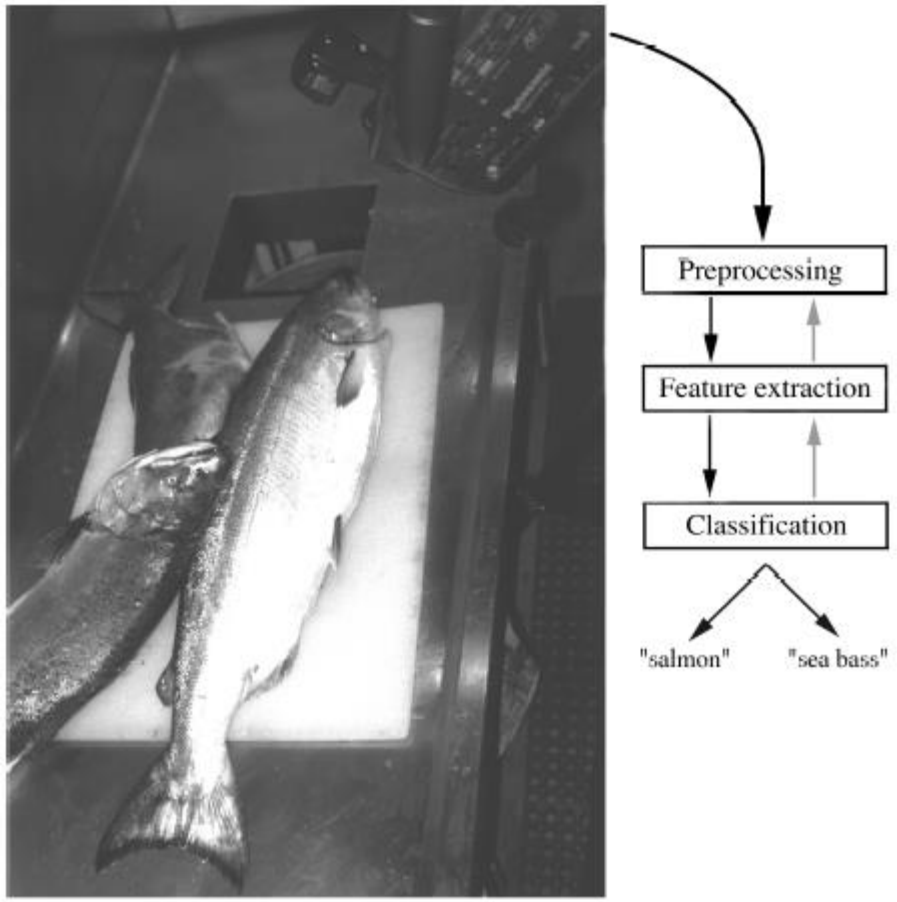
- Length, brightness, width, number and shape of fins, placement of the mouth, etc.
- Possible features for the classifier

Model

- Given that there are differences between salmon and sea bass, we assume that they can be described by different models (mathematical descriptions)

1. Create models for the different classes
2. Extract model parameters (features) from the data
3. Choose the model that the parameters correspond best to

Setup

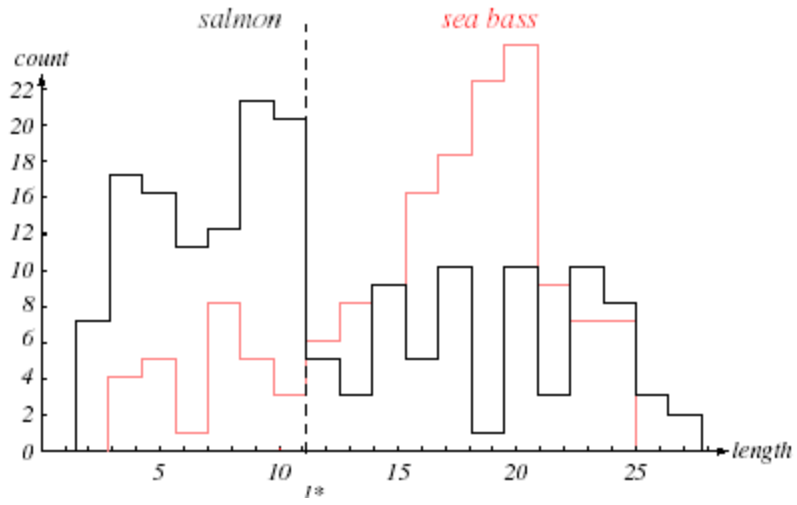


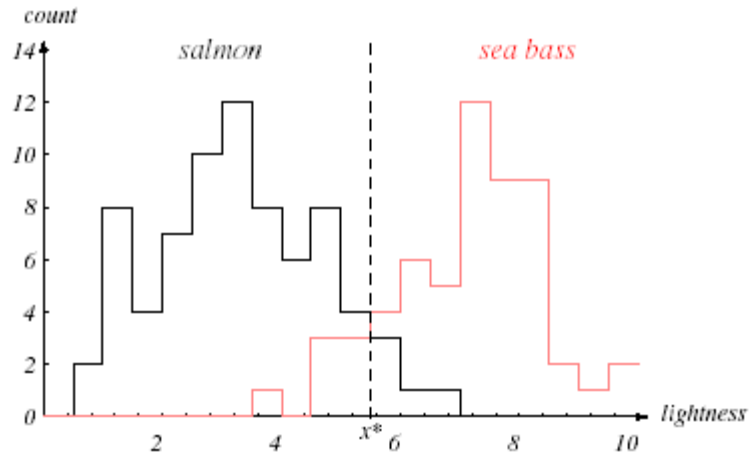
Preprocessing can involve adjusting image intensity or removal of the background.

Feature extraction

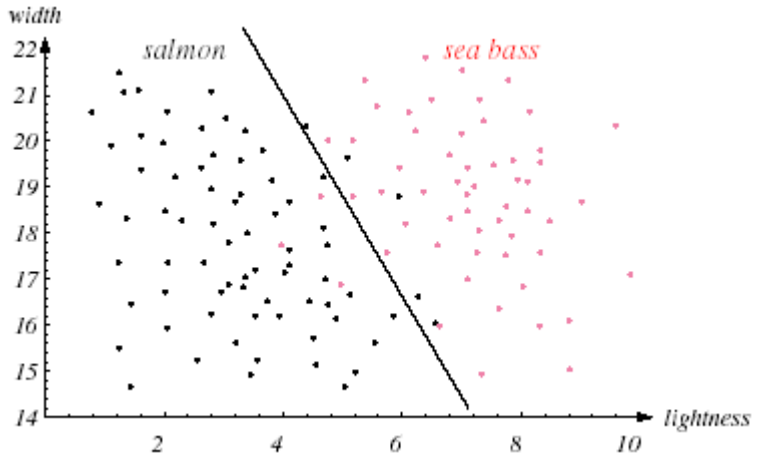
Reduce data by measuring features/values.

Measured features from a number of fish of both species can be used as training samples





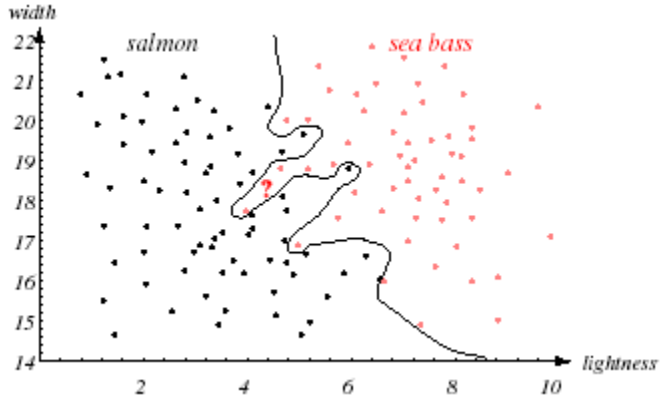
Multiple features – decison boundaries

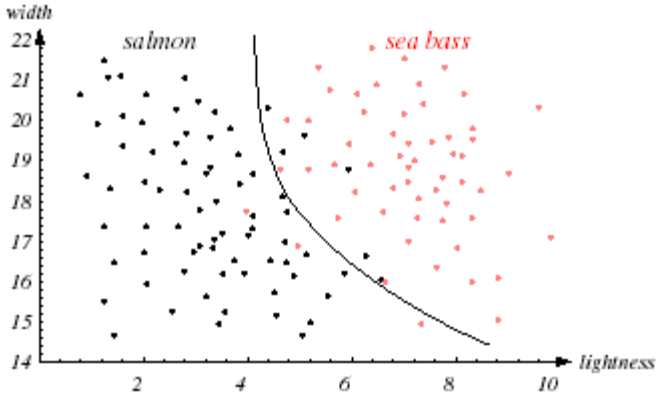


Straight lines are too simple when more complex models are needed

Perfect tuning based on trainingsamples is NOT the best solution

An optimal tradeoff between performance and simplicity of the classifier is preferable





Learning

Use of training data to learn or estimate the unknown parameters in the model

Supervised learning

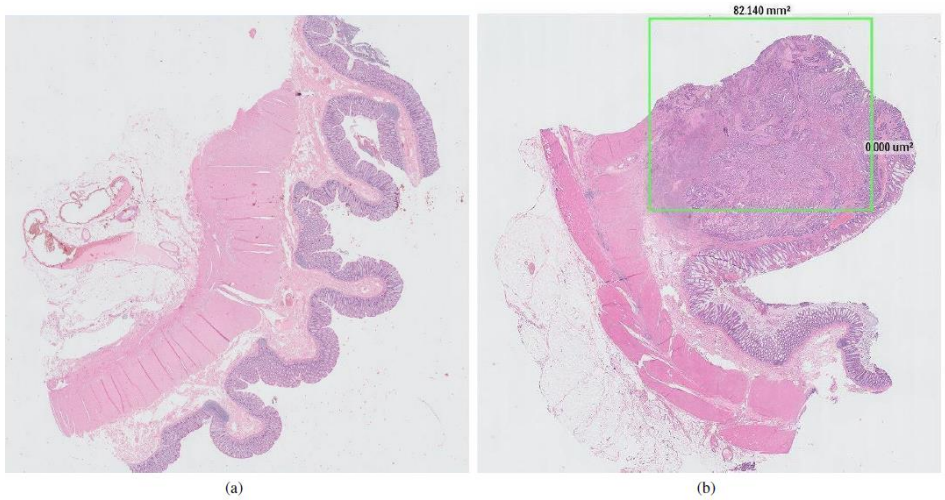
- The category/class is given by a ”trainer” for each pattern in a trainingset

Unsupervised learning

- There is no ”trainer”
- Clusters of ”natural grouping” are generated from input patterns

Digital pathology – detection of cancer

- Develop methods for automatic detection of cancer tissue within H&E stains
- Currently, manual inspection is used for the diagnosis
- Labor intensive and inter- and intra-observer variation
- Enable objective analysis of H&E stains

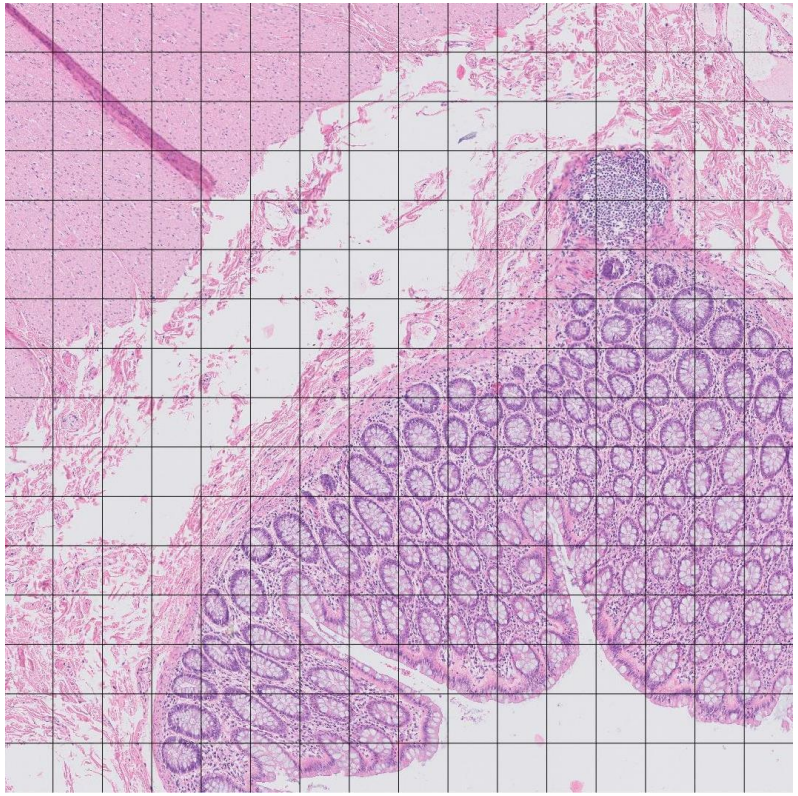


Normal colon

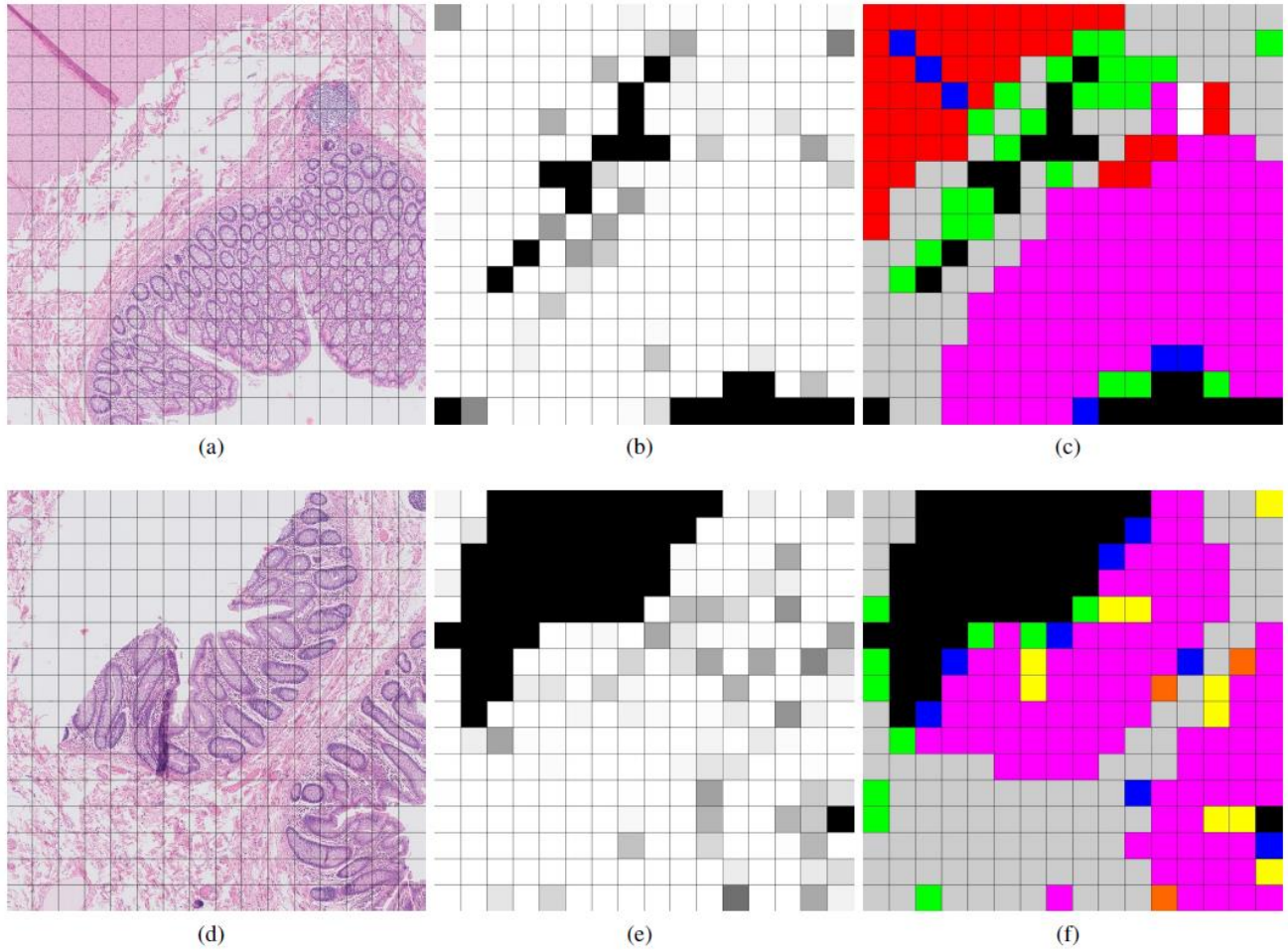
Colon adenocarcinoma

Patch based approach – Aim and motivation

- Small image regions (patches) can be used to represent different tissue subtypes
- Image patches allows to extract image information (features) from different tissue structures within whole slide H&E stains
- Use pattern recognition/machine learning to detect cancer tissue

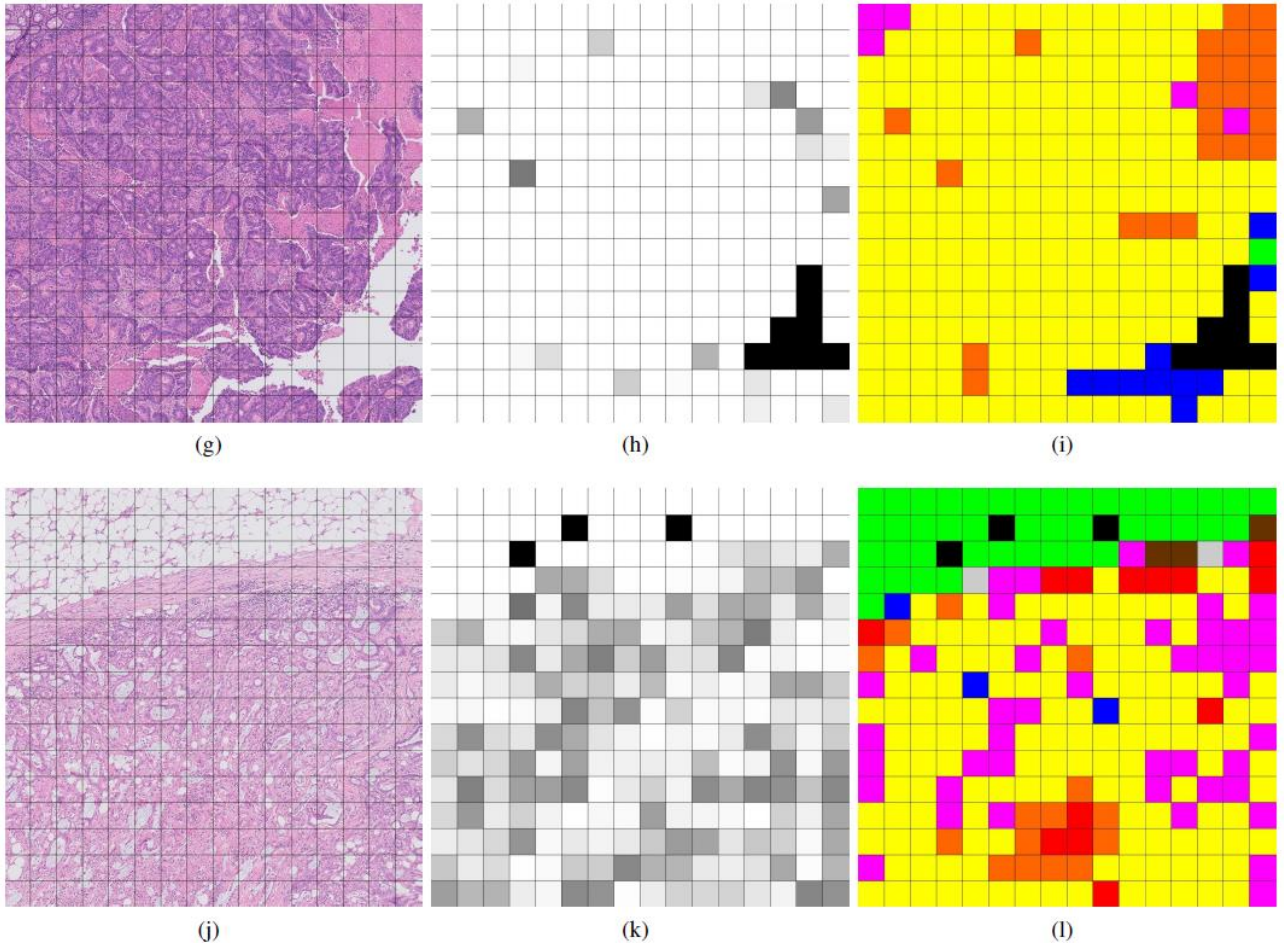


Colon cancer examples



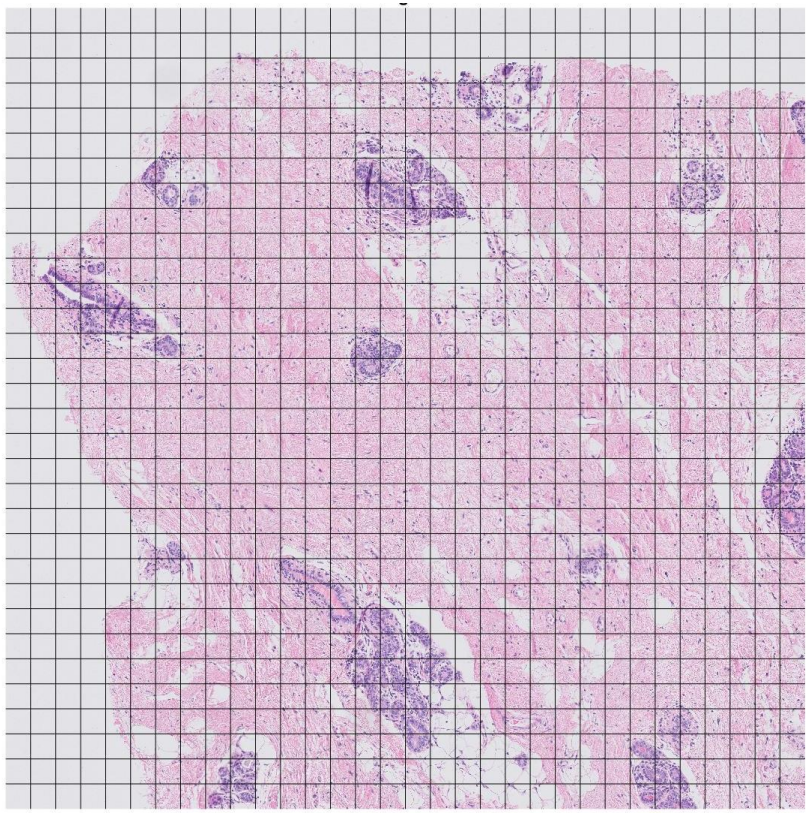
- yellow = cancer
- magenta = mucosa
- red = muscle
- green = fat
- blue = blood cells
- white = inflammation
- orange = necrosis
- brown = mucous
- gray = connective tissue
- black = excluded patches

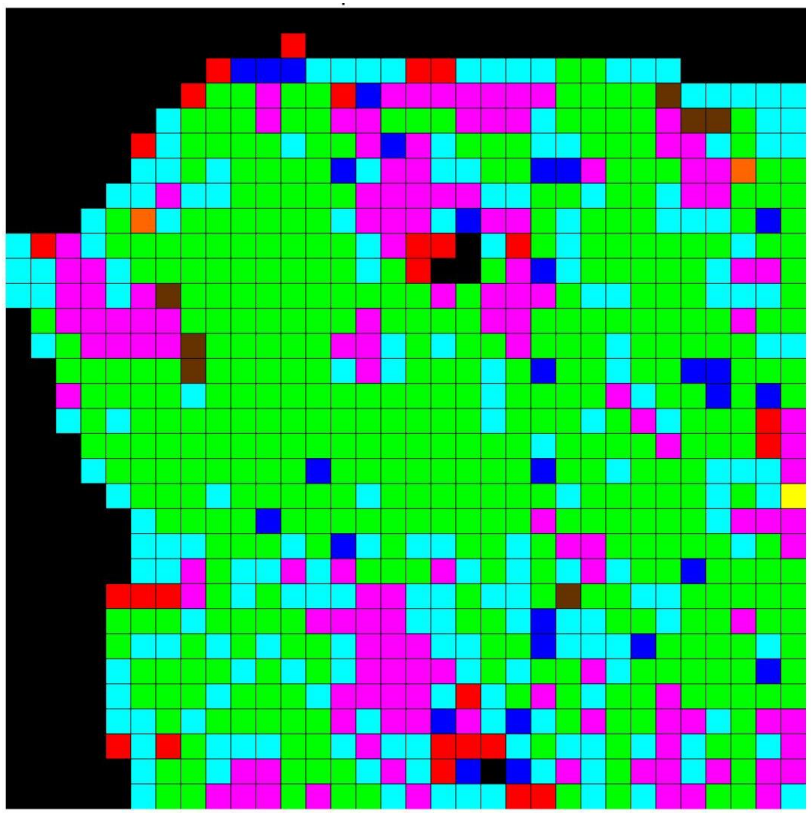
Colon cancer examples



- yellow = cancer
- magenta = mucosa
- red = muscle
- green = fat
- blue = blood cells
- white = inflammation
- orange = necrosis
- brown = mucous
- gray = connective tissue
- black = excluded patches

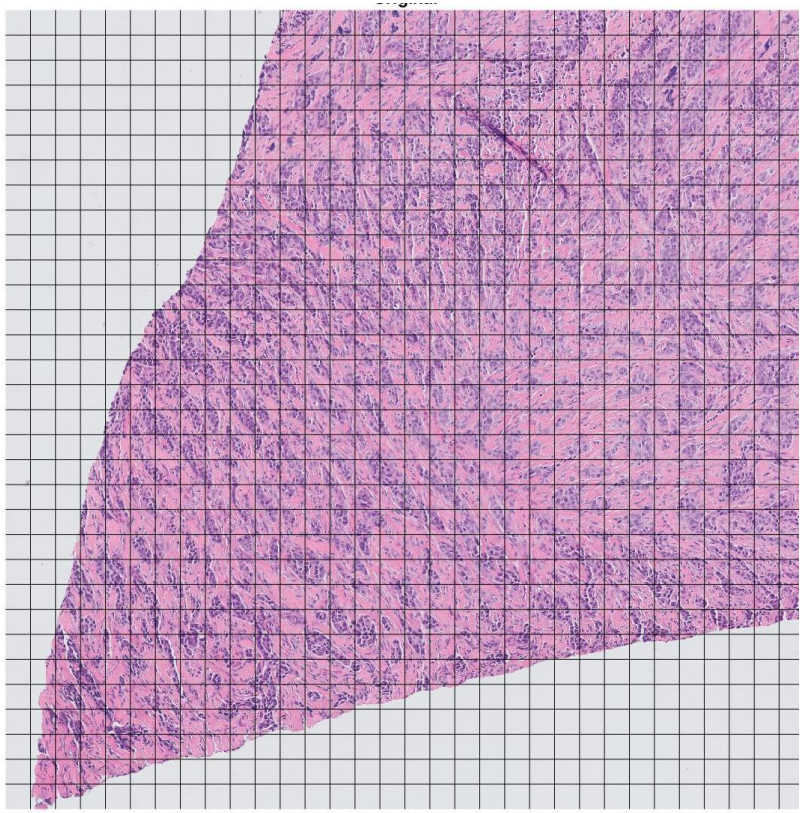
Breast cancer examples

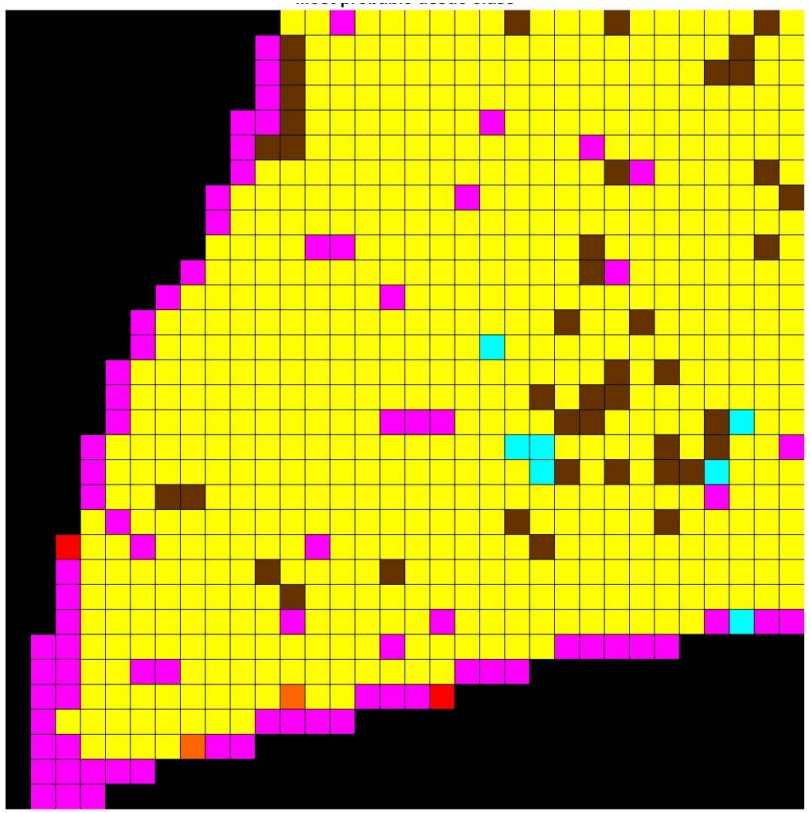




- yellow = cancer

Breast cancer examples





- yellow = cancer